

Generation Interconnection System Impact Study Report

For Queue Project Q21-1957

Asheville 138 kV

April 21, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1957. The Interconnection Customer proposes a 264.7 MW solar photovoltaic facility interconnecting at the Asheville 138 kV substation in the Virginia area. The request was received on December 15, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 605 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,909,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Nacoochee – South Texa 138kV
Contingency: Glen Oaks – South Texa 138kV
Loading: 127.6% (Rating: 659 MVA)
DFAX: 6.8%
Pre-project loading: 124.9%
2. Facility: Asheville – Ladysmith Hydro 138kV Ckt 1
Contingency: Ladysmith Hydro – Asheville 138kV Line
Loading: 117.5% (Rating: 722 MVA)
DFAX: 60.5%
Pre-project loading: 95.3%
3. Facility: Asheville – Ladysmith Hydro 138kV
Contingency: Ladymith Hydro – Asheville 138kV #1
Loading: 117.5% (Rating: 722 MVA)
DFAX: 60.5%
Pre-project loading: 95.3%
4. Facility: Ladysmith Hydro – West Bassett 138kV Circuit 1
Contingency: W Bassett – Ladysmith Hydro 138kV Cir. 2
Loading: 116.6% (Rating: 398 MVA)
DFAX: 37.4%
Pre-project loading: 91.7%
5. Facility: Ladysmith Hydro – West Bassett 138kV
Contingency: W Bassett – Ladysmith Hydro 138kV Line 1
Loading: 116.6% (Rating: 398 MVA)
DFAX: 37.4%
Pre-project loading: 91.7%
6. Facility: Leesville – Cloer 345kV L
Contingency: Clover – State School Substation 345kV
Loading: 114.9% (Rating: 1,643 MVA)

DFAX: 22.7%

Pre-project loading: 111.2%

7. Facility: Intel Substation – Suth Crk 138kV

Contingency: S Crk – Aibonito Pds Substation 138kV

Loading: 110.9% (Rating: 404 MVA)

DFAX: 5.1%

Pre-project loading: 107.5%

8. Facility: Hancock 345-138 kV TX

Contingency: Huntington Court 345 kV/138 kV Autotransformer

Loading: 107.6% (Rating: 415 MVA)

DFAX: 14.9%

Pre-project loading: 98.1%

9. Facility: Ladysmith Hydro – Leesville 138kV L

Contingency: Reusens – Huntington Court 345kV Line

Loading: 102.7% (Rating: 430 MVA)

DFAX: 24.9%

Pre-project loading: 87.4%

10. Facility: Desert Outpost – South Tea 138kV L

Contingency: Franklin-Elk Grove – South Texa Substation 138kV

Loading: 102.1% (Rating: 1,002 MVA)

DFAX: 7.6%

Pre-project loading: 100.1%

11. Facility: South Texa – Franklin-Elk Grove 138kV

Contingency: Desert Outost – South Texa 138kV

Loading: 101.6% (Rating: 774 MVA)

DFAX: 6.6%

Pre-project loading: 99.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Nacoochee – South Texa 138kV	Rebuild line with higher-capacity conductor	\$2,900,000	5% (\$149,000)
Asheville – Ladysmith Hydro 138kV Ckt 1	Rebuild line with higher-capacity conductor	\$8,704,000	15% (\$1,284,000)
Asheville – Ladysmith Hydro Substation 138kV Line 2	Rebuild line with higher-capacity conductor	\$8,769,000	15% (\$1,294,000)
Ladysmith Hydro – West Bassett 138kV Circuit 1	Rebuild line with higher-capacity conductor	\$12,926,000	15% (\$1,966,000)
Ladysmith Hydro – West Bassett 138kV	Rebuild line with higher-capacity conductor	\$9,961,000	15% (\$1,515,000)
Leesville – Clover 345kV line	Rebuild line with higher-capacity conductor	\$137,165,000	5% (\$6,351,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Intel Substation – Suth Crk 138kV	Rebuild line with higher-capacity conductor	\$8,958,000	5% (\$477,000)
Hancock 345-138 kV TX	Replace transformer with higher-rated unit	\$17,929,000	5% (\$860,000)
Ladysmith Hydro – Leesville 138kV L	Reconductor line	\$14,338,000	13% (\$1,809,000)
Desert Outpost – S Texa 138kV L	Reconductor line	\$761,000	3% (\$25,000)
South Texa – Franklin-Elk Grove 138kV	Reconductor line	\$5,125,000	4% (\$179,000)
	Total allocated to project		\$15,909,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.